FreshKeep Spatial

Your whole kitchen, at a glance — so nothing goes to waste



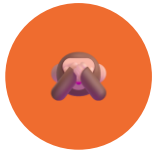
Abisheak S · IT21244384 · SE4041 Assignment 02 — Part B (visionOS)

Households throw away food they forgot they had

79 kg

of food wasted per person, per year, at home (UNEP Food Waste Index 2024)

Households are the single largest source of food waste.



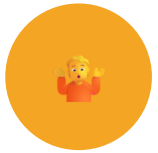
Out of sight, out of mind

Food at the back of the fridge is forgotten until it smells.



Tracking apps are tedious

Typing every item and date by hand — people give up in days.



No idea what to use first

Nothing tells you which food is about to expire today.

Why spatial computing?



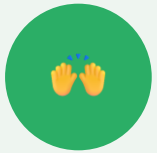
Kitchens are physical spaces

A 3D fridge you can look into beats a scrolling list.



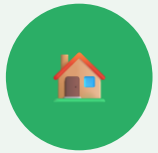
Glanceable urgency

Colour and distance show what expires first — no reading needed.



Hands-free while cooking

Look and pinch with messy hands; no phone to unlock.



Made for the home

Vision Pro is used at home, where the food decisions happen.

Three spatial experiences



Glass window

Pantry grouped by urgency, item details and recipe ideas. Ornament launches the spatial modes.



3D virtual fridge (volume)

Shelves for fridge, counter, pantry and freezer. Items are tokens coloured by freshness. Pinch to inspect; drag into the pot to cook.



Expiry timeline (immersive)

Food floats on an arc around you in your real kitchen — the closer it is, the sooner it expires.

Working prototype on visionOS

[visionos-main.png]

RealityKit volume

generated 3D shelves, tokens and pot — no asset files

Spatial gestures

SpatialTapGesture + DragGesture targeted to entities

Attachments

SwiftUI glass cards floating above 3D items

Shared data

same SwiftData model + FreshKeepCore package as iOS

Verified in CI

built and launched in the Vision Pro simulator on every push

Designed for eyes and hands



Look + pinch

Select an item; hover highlight shows what you are looking at



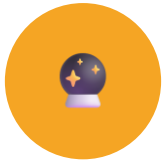
Pinch + drag

Pick food up and drop it into the Cook-now pot to log it as eaten



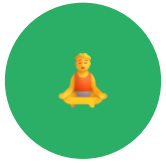
Ornaments

Primary actions float below the window, following Apple HIG



Glass & depth

System glass materials keep the room visible and comfortable



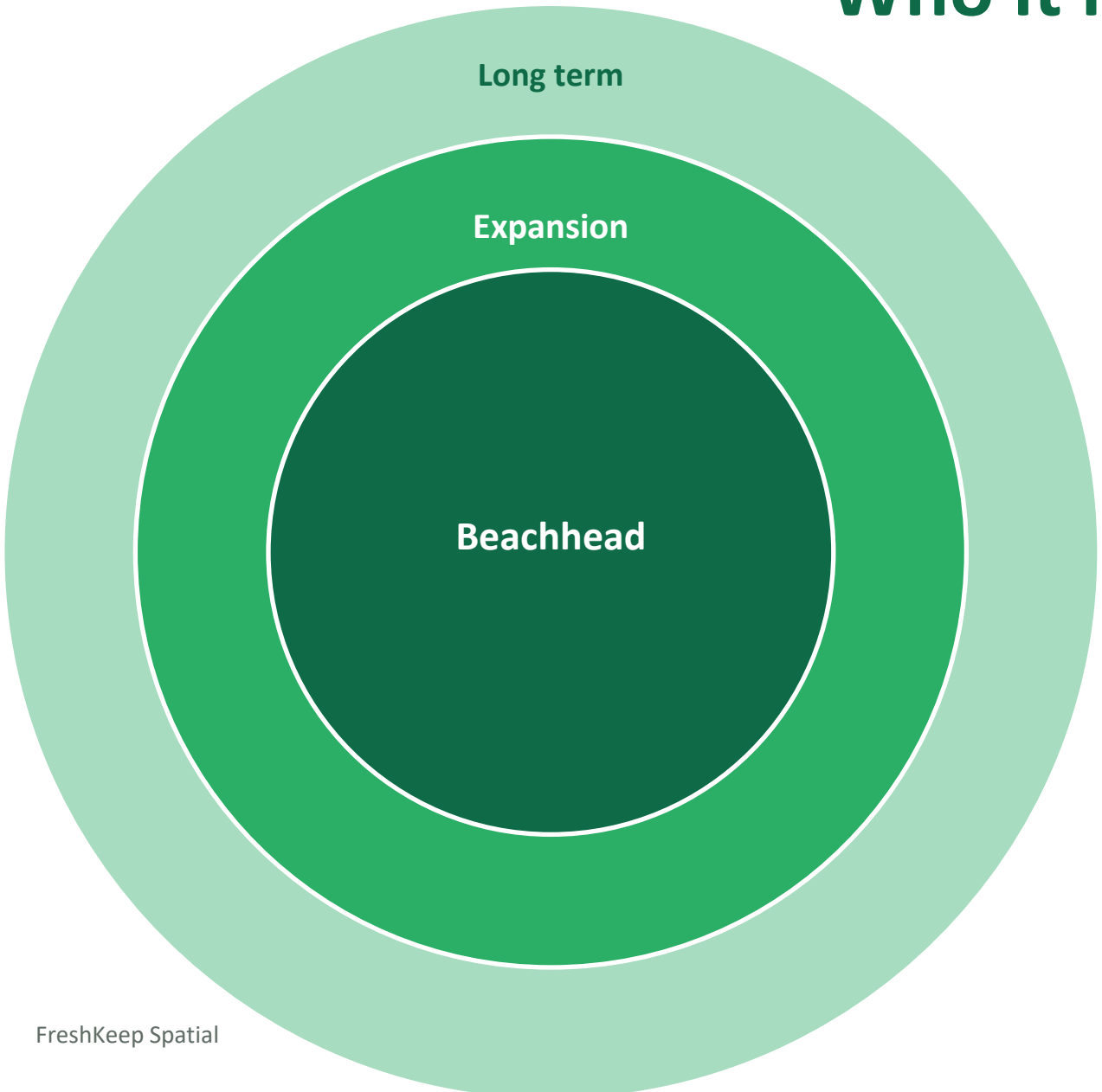
Comfort

Content at 0.8 – 2.2 m, mixed immersion, no forced motion

Existing apps solve only part of it

Product	Platform	What it does	Gap
Crouton	iOS, visionOS	Recipe manager and cooking mode	No pantry or expiry awareness
Lowe's Style Studio	visionOS	Design a kitchen spatially	Kitchen layout, not food
Smart fridges with cameras	Appliance	Look inside the fridge remotely	Expensive hardware; fridge only
NoWaste / Fridgely	iOS	Manual inventory and expiry lists	Typing; not spatial
FreshKeep Spatial	iOS + visionOS	AI capture on iPhone, 3D pantry & timeline on Vision Pro	**Whole kitchen, urgency-first**

Who it is for



Beachhead

FreshKeep iPhone users who own an Apple Vision Pro — early adopters, home cooks

Expansion

All Vision Pro households; families who share a kitchen

Long term

Lighter, cheaper headsets and smart-home partners

Revenue model: freemium + partnerships

Free

iPhone app

- Scan food & labels
- Reminders
- Insights

FreshKeep Plus

USD 2.99 / month · USD 24.99 / year (proposed)

- Spatial fridge & timeline
- Household sharing & iCloud sync
- Premium recipe packs

Partners

B2B (future)

- Grocery partners: “use it up” offers
- Appliance makers: licensed spatial view
- Anonymised waste insights for NGOs

Go-to-market plan

1

Launch iOS

Free on the App Store; campus and sustainability communities in Sri Lanka

2

Add Spatial

Release FreshKeep Plus with the visionOS app; App Store feature pitch

3

Build proof

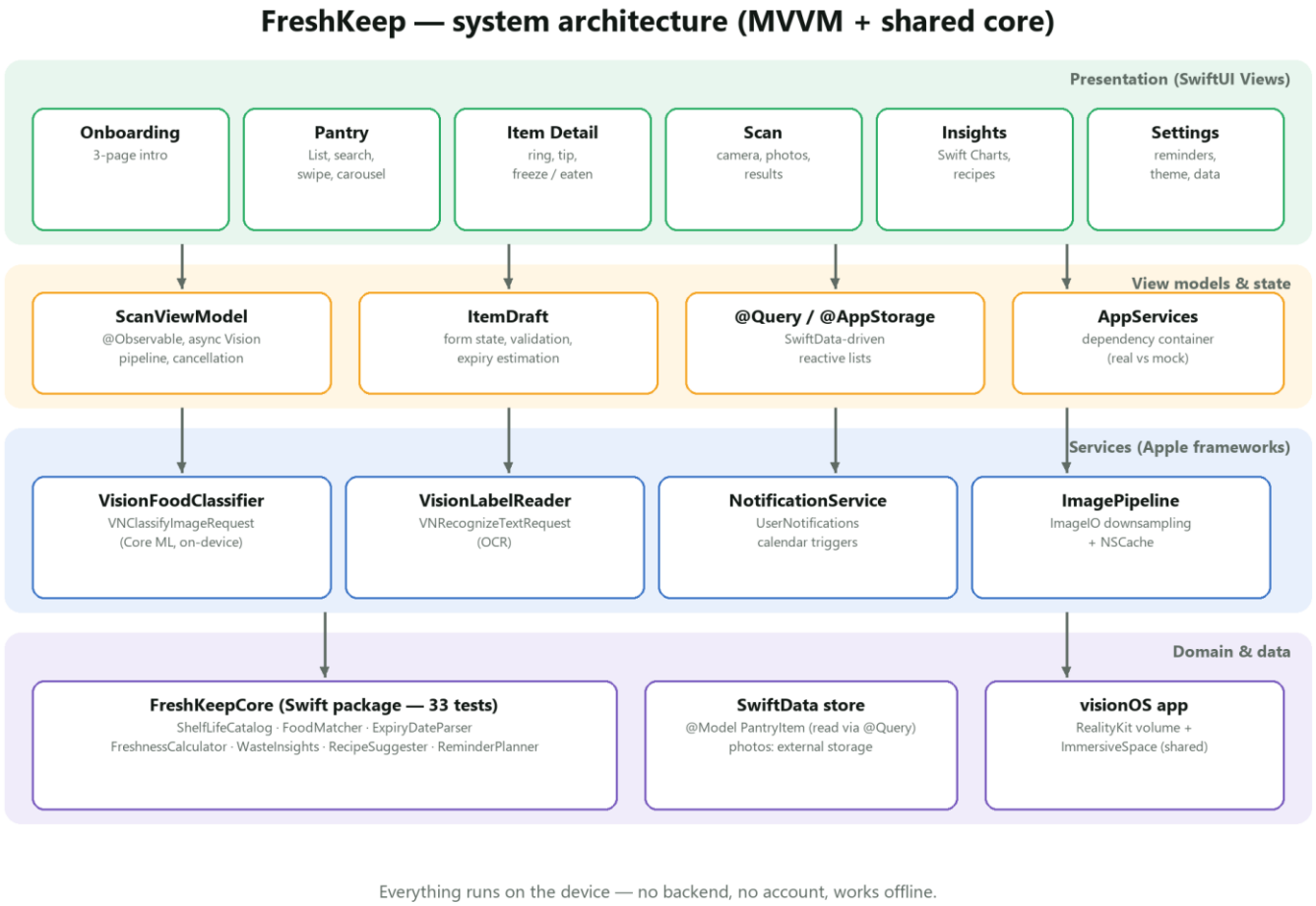
Publish rescue stats; partner with food-waste NGOs and cooking creators

4

Partner

Pilot "use it up" offers with grocers; explore appliance licensing

Built to scale across Apple platforms



Everything runs on the device — no backend, no account, works offline.

- One Swift package shared by iPhone and Vision Pro
- SwiftData model shared by both apps
- On-device Vision AI — private and offline
- RealityKit entities generated in code (small app size)
- 50+ automated tests run on every push

Impact and roadmap

Less waste

Urgency-first view and reminders target the main cause: forgetting

Money saved

Every rescued item is tracked in LKR so users see the benefit

Lower emissions

Rescued food avoids its CO_2e footprint

Next

- Anchor labels to the real fridge with object tracking
- SharePlay household mode: plan meals together
- Custom Core ML model for local produce (jackfruit, gotukola)
- Apple TV / iPad kitchen dashboards

See your food. Save your food.

Thank you — questions? · github.com/IT21244384/SE4041-FreshKeep

